Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/16/2004
PI Name:	DeTitta, George T. Ph.D.		
Project Title:	Macromolecular Crystallization: Physical Principles, Passive Devices, and Optimal Protocols		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Buffalo	**State:**	NY
Zip Code:	14203-1196	**Congressional District:**	28
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Luft, J. (Hauptman-Woodward Medical Research Institute)		
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The work centers on developing an understanding of macromolecular crystallization and improving available methods for crystallization. We have proposed and fabricated a number of devices and methods that can work to control the kinetics of equilibration in vapor diffusion crystallization experiments in an effort to improve crystal quality. Our work with microbatch crystallization has allowed us to exploit temperature and greatly improve the efficiency of the experiments. It has also allowed us to develop high throughput methods of conducting macromolecular crystallization in 1536 well plates. These methods have allowed us to set up an unprecedented number of crystallization experiments and collect data that is being used to improve our understanding of the crystallization process. Our goal is to understand the importance of individual crystallization parameters to an experiment's outcome: develop efficient, passive methods that will exploit these variables and lead to the production of high quality crystals. The objective of our work is to design, test, and evaluate devices and protocols for growing single crystals of macromolecules suitable for X-ray diffraction. In the

Task Description:	process we propose to identify those devices and protocols that will work optimally on earth, and to identify those that may benefit from a microgravity environment. Our objective is to quantitatively assess those parameters that have a profound effect on crystal quality. Macromolecular crystallography provides the structural data that forms a foundation for understanding molecular processes. Protein engineering, molecular genetics, and rational drug design depend on an accurate description of the structure of a macromolecule (typically a protein) at the atomic level. Such a view is afforded by few disciplines, and the principal one is crystallography. A necessary first step in structure determination is the growth of a single crystal suitable for diffraction studies. Our work is aimed at a greater understanding of the processes of crystal growth, and at the design and testing of devices and protocols to more efficiently grow and improve the quality of macromolecular crystals. Modern approaches to drug design rely on an accurate understanding of the target molecules of drugs and therefore our work has the potential to materially affect the general public who will benefit from new drugs that result from these approaches.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A high throughput crystallization facility was established at the Hauptman-Woodward Institute in February 2000. The laboratory uses Tango liquid handling systems to set up microbatch under oil crystallization experiments in 1536 well microassay plates. The liquid handling systems are used to reformat from low density to high density plates using source/destination plate protocols. Results of the experiments are recorded over a one month time course using an automated reader table. A significant computer infrastructure has been established for the facility. The infrastructure has layers of redundant systems and failover mechanisms to prevent disruption of data collection by a single point of failure. A central database monitors and controls many aspects of the laboratory including optical disc and tape backup, sample status email notification to users of the facility, and control of the ftp server used to transfer images to outside users. Notification that images are available by ftp takes place minutes after a plate has been recorded. This provides outside users with nearly instantaneous access to the results of their experiments. As of April 2004 the lab was used to execute 4.8 million crystallization experiments on 3100 samples submitted by more than 500 investigators. More than 33 million digital images are currently archived, showing the results of each experiment over a one month time course. These experiments are being used as a source of data to look for trends in crystallization behavior. The facility is also being used as a tool to conduct basic experiments on crystallization. The method of using dyes to identify macromolecular crystals is not a novel one and in fact dates back to 1898 (F.G. Hopkins and S.N. Pinkus (1898) J. Physiol. 23, 130-136). Using a HTS method we were able to investigate a series of 90 different compounds for their ability to validate protein crystals. The verification method can be utilized in manual or high-throughput modes. Our goal was to identify a compound that reliability identified whether a crystal is composed of an inorganic salt or macromolecule under the diverse chemical environments encountered in macromolecular crystallization experiments. This tool will greatly enhance image analysis efforts. It will also improve the efficiency of optimization by eliminating conditions that produce salt crystals from the crystallization leads used for optimization trials. We have identified compounds, both anionic and cationic, that effectively identified macromolecular crystals. The most effective compound identified thus far, the mono-azo dye Chromotrope 2B, is being evaluated to determine its effectiveness with a variety of macromolecular crystals. The commercially available standard for dyeing crystals, methylene blue was also shown to be quite effective in our testing. The concept of using a precipitation reaction index to be predictive of macromolecular crystallization conditions is under development in collaboration with Dr. Igor Jurisica, Ontario Cancer Institute, Toronto, ON. The central hypothesis is that similar proteins will crystallize under nearly identical conditions. We are using a series of precipitation reactions as a quantitative measure of protein similarity. We developed 1536 unique precipitation cocktails to be screened against each protein. After setting up and imaging the precipitation reaction using the high throughput crystallization facility, the results are scored and encoded. The encoded precipitation reactions for proteins that we know how to crystallize (knowns) are organized in a database. If our hypothesis is correct, we should be able to measure the code for a protein that we do not know how to crystallize (unknown) and compare it to the database codes. If an unknown protein code has high correlation with that of a known protein in the database, we believe the crystallization conditions for the known will provide good lead conditions for crystallization of the unknown protein. Automated image analysis is critical to the success of this project and we are collaborating with several groups as well as working on our own in-house system for the automated classification of the images. A large scale manual image classification of 150,000 images from 96 protein samples is underway. Truth data generated by this classification study will be used to refine automated scoring software as well as search for trends in crystallization behavior. The classified images will be used in conjunction with measured properties of the crystallization cocktails (density, surface tension, vapor pressure, viscosity, and pH) with a goal of being able to predict crystallization conditions for macromolecules. Several important observations have come out of our initial analysis of the results of these crystallization trials from a diverse selection of 3100 macromolecular samples. The benefits of using 1536 cocktails is clear. The cocktails offer a wide coverage of chemical conditions during the screening trials (although it is still a very coarse screen of the actual conditions). Multiple lead conditions for crystallization (when they appear) often produce crystals with varying physical properties (space groups, molecular packing arrangements, solvent channel dimensions and diffraction limits). This can be extremely advantageous to an individual investigator undertaking a particular study. Oftentimes, we will only observe a few leads or a single lead. These leads often come from chemcial conditions that are outside of the routine chemical screening parameters, but are included in the 1536 cocktails. Finally, we are able to set up 1536 unique crystallization experiments using only 4mg of protein in less than ten minutes. This reduces the time for sample degradation when compared with manual screening.
Bibliography Type:	Description: (Last Updated: 03/26/2008)
Articles in Peer-reviewed Journals	Cumbaa CA, Lauricella A, Fehrman N, Veatch C, Collins R, Luft J, DeTitta G, Jurisica I. "Automatic classification of sub-microlitre protein-crystallization trials in 1536-well plates." Acta Crystallogr D Biol Crystallogr. 2003 Sep;59(Pt 9):1619-27. PMID: 12925793 , Sep-2003

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